

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find $f'(2)$ for $f(x) = \frac{1}{x+3}$.

$$\begin{aligned}
 \text{Solution: } f'(2) &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{1}{(2+h)+3} - \frac{1}{2+3}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{1}{5+h} - \frac{1}{5}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{5 - (5+h)}{(5+h) \cdot 5 \cdot h} \\
 &= \lim_{h \rightarrow 0} \frac{-h}{(5+h) \cdot 5 \cdot h} \\
 &= -\frac{1}{25}
 \end{aligned}$$

Problem 1. (5 points) If $f(x) = 2x^2 - 1$, find $f'(x)$ and $f''(x)$.

$$\begin{aligned}
 \text{Solution: } f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{[2(x+h)^2 - 1] - [2x^2 - 1]}{h} = \lim_{h \rightarrow 0} \frac{[2x^2 + 4xh + 2h^2] - [2x^2 - 1]}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h} = \lim_{h \rightarrow 0} (4x + 2h) \\
 &= 4x \\
 f''(x) &= \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4(x+h) - 4x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4h}{h} = 4
 \end{aligned}$$